

Intro:

So far my environment is able to train an LLM with a verifiable reward for crystal relaxation/generation and Reinforcement Learning (GRPO). Is a nice approach cause it promotes reasoning from the LLM and eventually the LLM learns "supposedly" actual chemistry and physics to climb the reward.

The reward design is a mix of:

- Parsing (can i even read the CIF file with pymatgen package?)
- Bond Heuristics (measures if bonds/distances between atoms are between 32% of their theoretical values)
- Formation Energy (Using CHGNet graph neural network) below $<0.1\text{eV/atom}$
- Forces (also with CHGnet) have to be close to net zero

So far this approach worked. This is not an easy task, so just starting from models (not frontier ones) it is hard to get a reward signal. But many things can either be improved or extended.

First, Agentic AI is now focusing on the best approach to solve Long tasks (on the orders of days/weeks) so the current limitation that DFT takes forever may make it suitable to use.

The goals I have are:

- Being able to realistically run DFT simulations of generated crystals on Virtual Machines/sandboxes. Why? Because is the only way that this can be useful at large scale training runs where im probably going to be sending a hundred different simulations.
- Make sure, up to a reasonable degree, that the simulations are scientifically useful. With my experience in MD I know a simulation can "run" and being completely worthless if you're parameters are incorrectly tuned.

What is nice about this? Suddenly my goal can become way more challenging that just "create a stable material", but can even be something like: Based on the closely related structures how is this Structure stable against the convex hull.

If we're able to just get a stable signal using DFT for "any" Structure, this idea is worth a million dollars. But of course, we can start just testing it on known Structures or families. And try to generalize later.